

BPL_TEST2_Batch - demo

In [1]: *# Import the application model and context variables*
`from BPL_TEST2_Batch import*`

Windows - run FMU pre-compiled JModelica 2.14

In [2]: *# Import the general FMU_explore functions ana adapt them to the application*
`from fmu_explore_pyfmi import fmu_explore`

`Dummy = fmu_explore(model, parValue, parLocation, parCheck, fmu_model, fmu_proc
MSL_usage, MSL_version, BPL_version, \
opts_std, simulationTime, timeDiscreteStates, stateValue, \
diagrams, ax, lines)`

`par = Dummy.par
init = Dummy.init
disp = Dummy.disp
simu = Dummy.simu
show = Dummy.show
resetPen = Dummy.resetPen
process_diagram = Dummy.process_diagram
describe_general = Dummy.describe_general
describe_parts = Dummy.describe_parts
describe_MSL = Dummy.describe_MSL
FMU_explore_info = Dummy.FMU_explore_info
system_info = Dummy.system_info
SDG = Dummy.SDG`

In [3]: *# Bring in the user made fmu_explore functions newplot() and describe()*

In [4]: `run -i BPL_TEST2_Batch_explore.py`

Model for the process has been setup. Key commands:

- `par()` - change of parameters and initial values
- `init()` - change initial values only
- `simu()` - simulate and plot
- `newplot()` - make a new plot
- `show()` - show plot from previous simulation
- `disp()` - display parameters and initial values from the last simulation
- `describe()` - describe culture, broth, parameters, variables with values/units

Note that both `disp()` and `describe()` takes values from the last simulation and the command `process_diagram()` brings up the main configuration

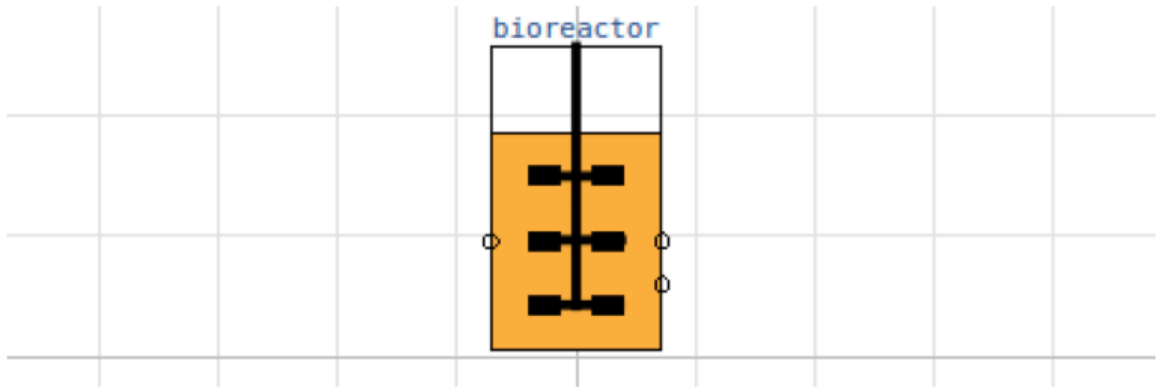
Brief information about a command by `help()`, eg `help(simu)`

Key system information is listed with the command `system_info()`

In [5]: `%matplotlib inline
plt.rcParams['figure.figsize'] = [25/2.54, 20/2.54]`

In [6]: `process_diagram()`

No processDiagram.png file in the FMU, but try the file on disk.



```
In [7]: describe('process')
```

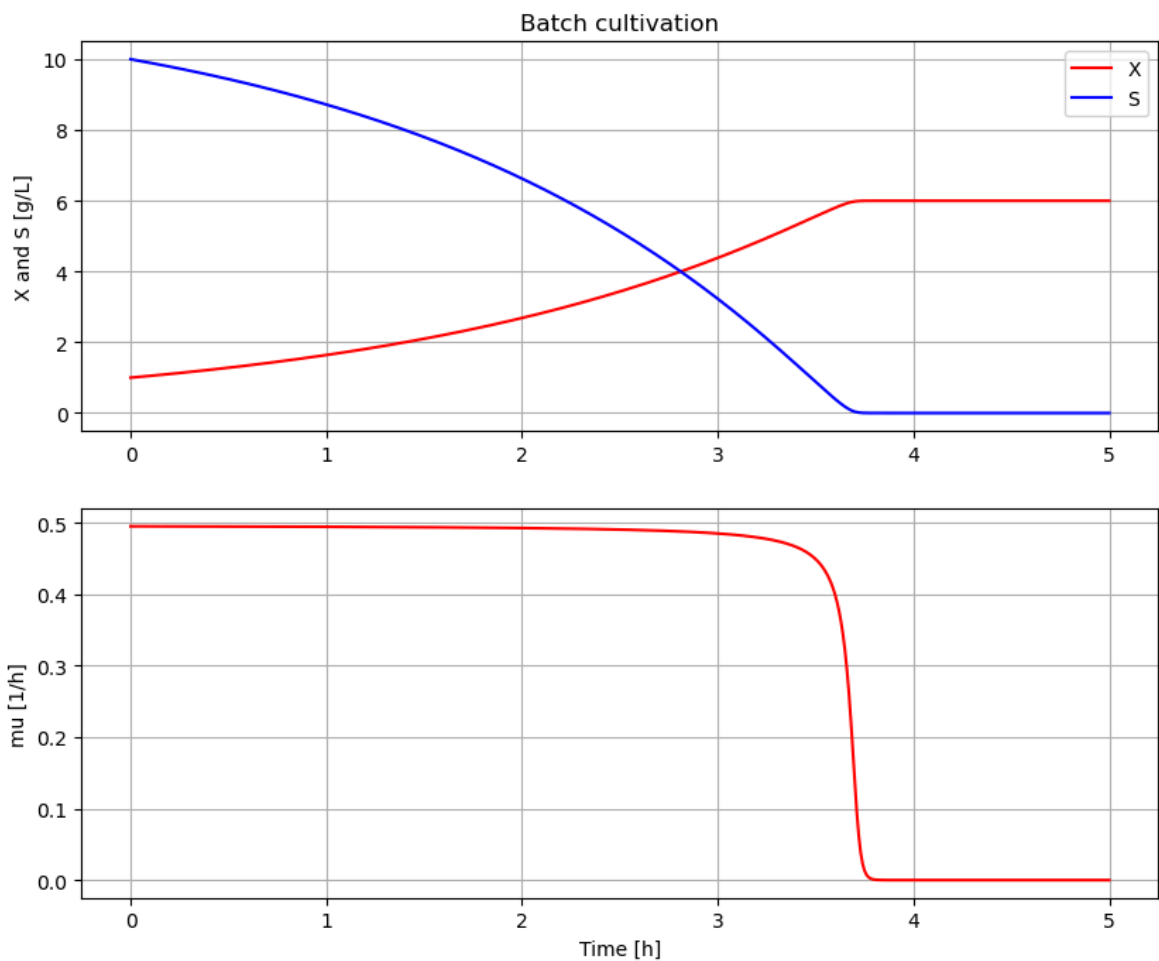
```
In [8]: describe('culture')
```

Simplified text book model - only substrate S and cell concentration X

```
disp('bioreactor', mode='long')
```

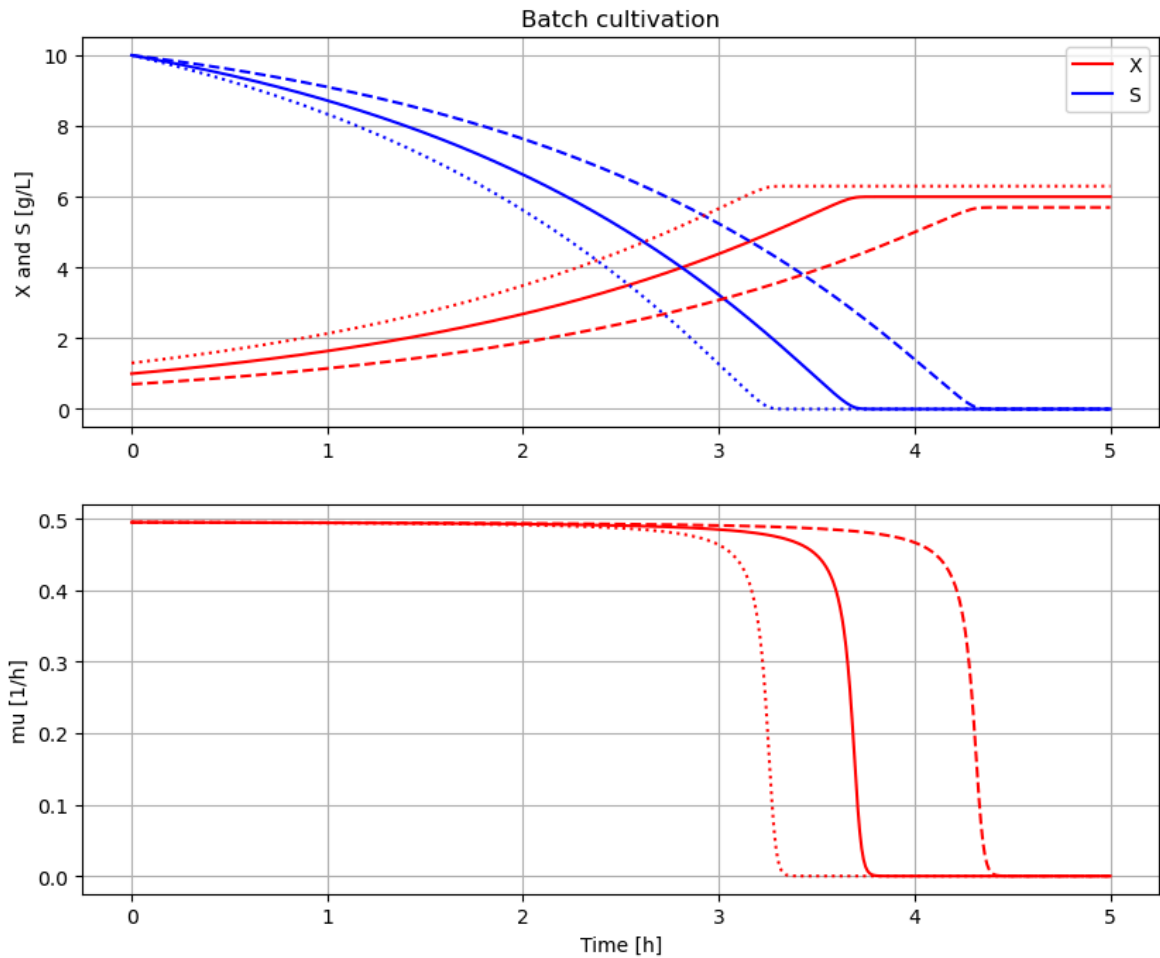
```
In [9]: # Simulation with default values of the process
newplot(plotType='TimeSeries')
simu()
```

```
Out[9]: <pyfmi.fmi_algorithm_drivers.FMIResult at 0x264cee91c70>
```



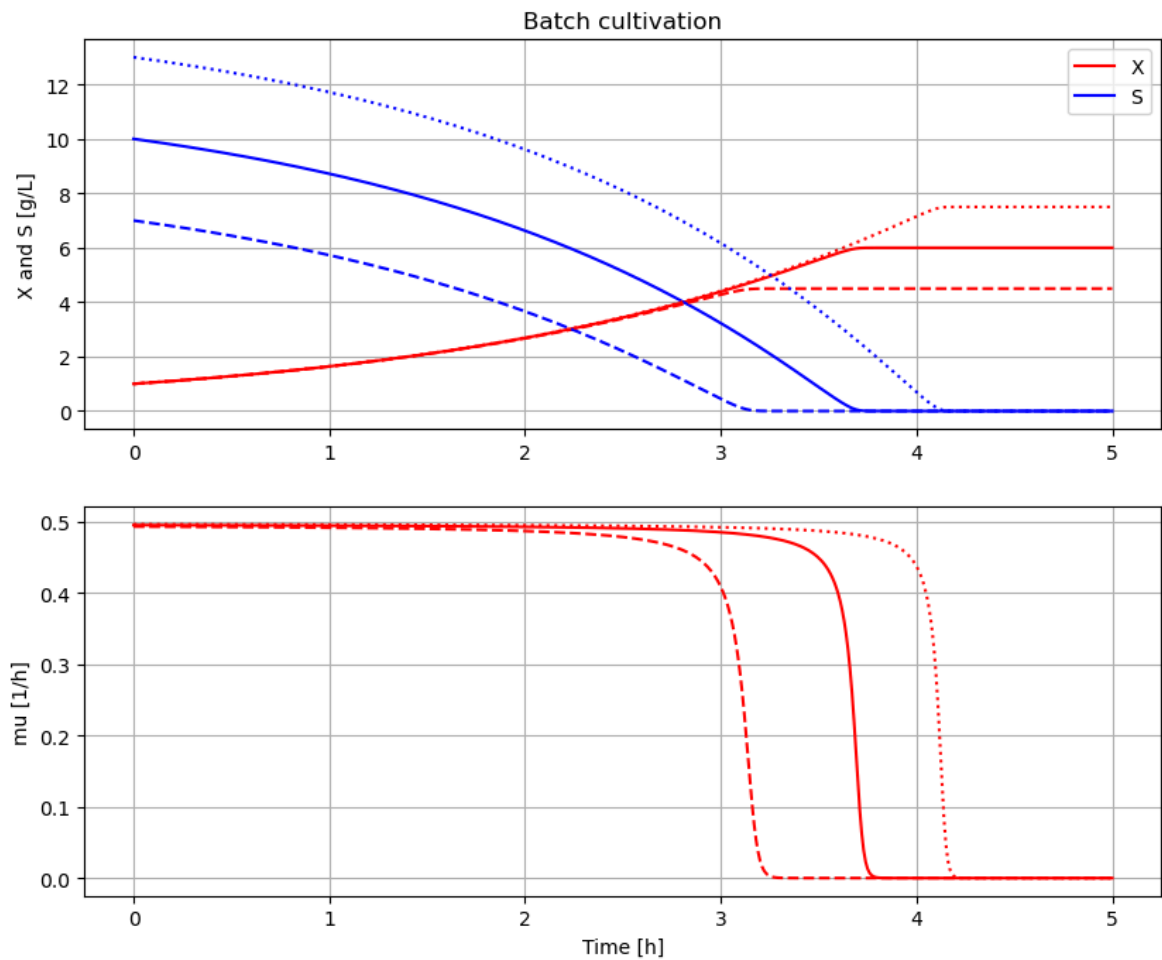
```
In [10]: # Simulation were initial value of biomass VX_0 is varied
newplot(plotType='TimeSeries')
for value in [1.0, 0.7, 1.3]: init(VX_start=value); simu(5)
```

```
# Restore default value of VX_0
init(VX_start=1.0)
```



```
In [11]: # Simulation were initial value of substrate VS_start is varied
newplot(plotType='TimeSeries')
for value in [10, 7, 13]: init(VS_start=value); simu(5)

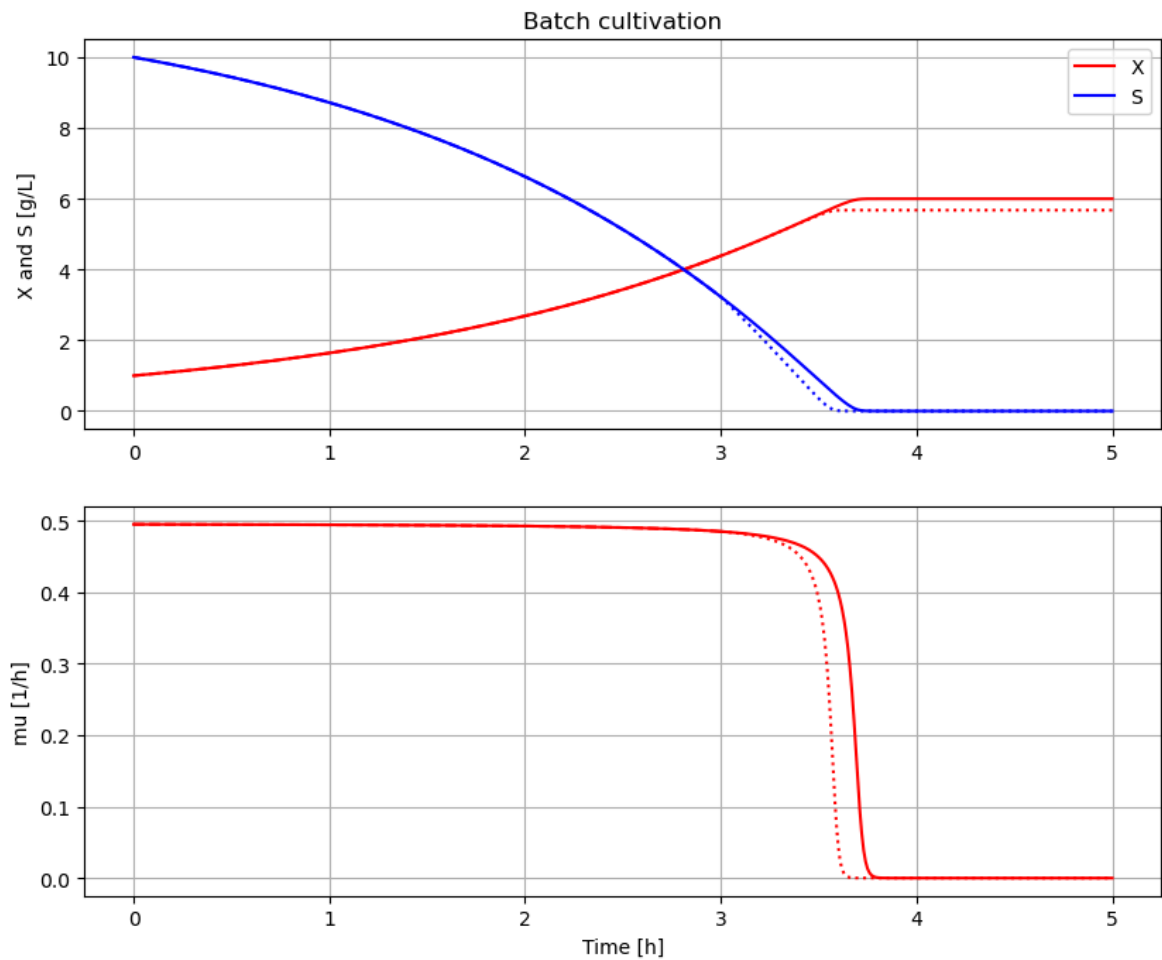
# Restore default value of VS_start
init(VS_start=10)
```



```
In [12]: # Simulation where metabolism is changed after 3 hours
newplot(plotType='TimeSeries')
simu(5)

simu(3)
par(Y=0.4, qSmax=1.0/(0.4/0.5)); simu(2, 'cont')

# Restore default value of Y and qSmax
par(Y=0.5, qSmax=1.0)
```



```
In [13]: disp('culture')
```

```
Y : 0.4
qSmax : 1.25
Ks : 0.1
```

```
In [14]: describe('mu')
```

```
Cell specific growth rate variable : 0.0 [ 1/h ]
```

```
In [15]: describe('parts')
```

```
['bioreactor', 'bioreactor.culture', 'MSL']
```

```
In [16]: describe('MSL')
```

```
MSL: none
```

```
In [17]: system_info()
```

System information

- OS: Windows
- Python: 3.12.11
- Scipy: not installed in the notebook
- PyFMI: 2.21.0
- FMU by: JModelica.org
- FMI: 2.0
- Type: FMUModelCS2
- Name: BPL.Examples_TEST2.Batch
- Generated: 2026-04-09T15:38:29
- MSL: 3.2.2 build 3
- Description: Bioprocess Library version 2.3.2
- Interaction: FMU-explore version 1.1.3

In [18]: `SDG('explan')`

"Soli Deo Gloria"

It is latin and means "To the honour of God".

The great composer Johan Sebastian Bach used to end his compositions with this small remark SDG.

And I like to do that too :).

In []: